

GREAT SCIENTISTS

Leonardo da Vinci

The Italian artist Leonardo da Vinci (1452–1519) was also an extremely clever scientist. As well as painting the *Mona Lisa*, one of the most famous works of art of all time, he studied anatomy, geology, geography, and optics. He was a brilliant engineer and drew designs for submarines, parachutes, and airships centuries before the technology existed to build them.

Renaissance Florence

In 14th-century Italy, people began to take a fresh interest in Greek and Roman learning that had been lost for centuries. By the late 15th century, the city of Florence, Leonardo's birthplace, was at the heart of what is called the Renaissance (rebirth), a time of cultural renewal. From art to medicine and from architecture to engineering, scholars relearned old techniques and discovered new ones.

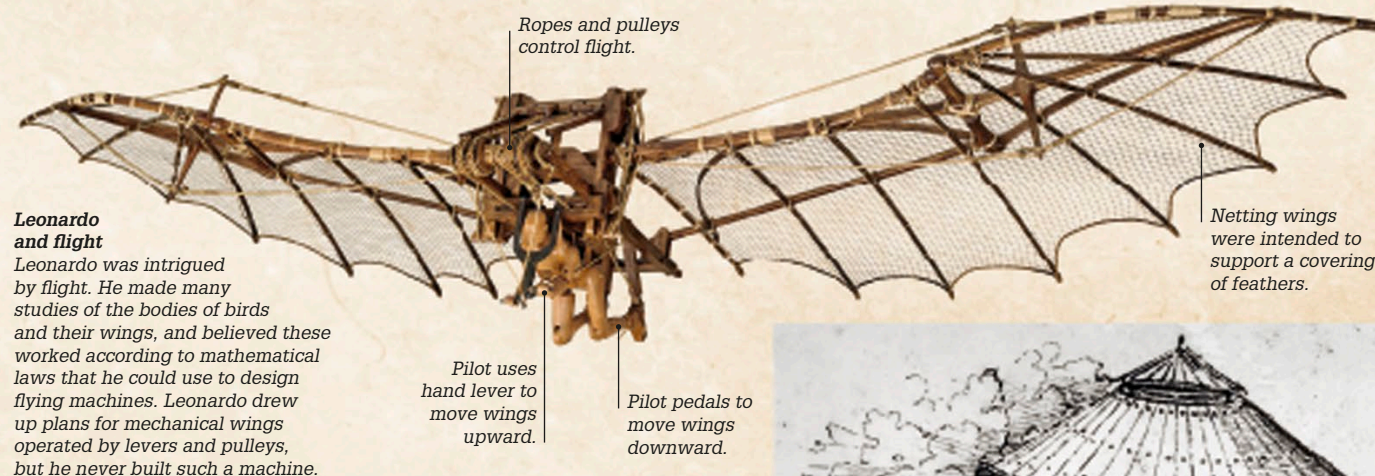
The engineer

Leonardo's desire to understand how things worked, combined with his skill at technical drawing, sparked in him an interest in machines and engineering. He designed complex levers, pulleys, and springs for use in construction. Leonardo was also a talented military engineer, and in 1500, he advised the Venetians on how to defend themselves from attacks by the Turks. One of his suggestions was to use a form of submarine to sink enemy ships. In his lifetime, Leonardo's inventions attracted little public interest. Today, we recognize his importance to science.



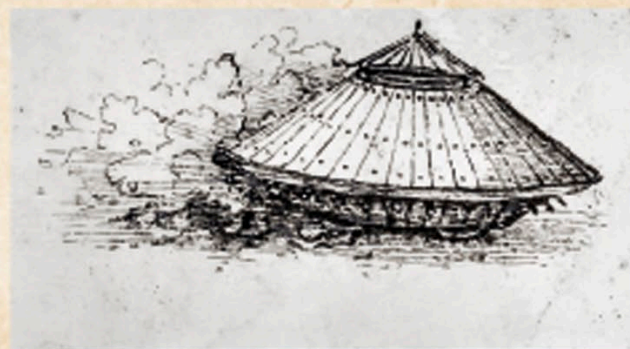
The anatomist

From the 1490s, Leonardo studied anatomy. He dissected animals and attended post-mortems of human corpses so that he could see the internal structure of the body. As a result, he was able to produce a series of highly detailed anatomical sketches.



Leonardo and flight

Leonardo was intrigued by flight. He made many studies of the bodies of birds and their wings, and believed these worked according to mathematical laws that he could use to design flying machines. Leonardo drew up plans for mechanical wings operated by levers and pulleys, but he never built such a machine.



Leonardo's inventions

This wooden tanklike vehicle for storming fortifications was just one among many ingenious machines that Leonardo devised. Others included a parachute, a dredging machine, and a robotic knight that could grasp objects and open and close its jaw.

“No human investigation can be called real science if it cannot be demonstrated mathematically.”

Leonardo da Vinci, *Trattato della Pittura*
(Treatise on Painting)